BIOL 110 – Lecture 4: Photosynthesis

Photosynthesis converts light energy into chemical energy stored in glucose. It takes place in the chloroplasts of plant and algal cells.

Light-dependent reactions

These occur in the thylakoid membranes. Chlorophyll absorbs mainly blue and red light. Water is split (photolysis), releasing oxygen as a by-product, and the energy is captured as ATP and NADPH.

Calvin cycle (light-independent reactions)

In the stroma, the enzyme RuBisCO fixes carbon dioxide onto RuBP. ATP and NADPH from the light-dependent reactions are used to make G3P, which builds glucose.

Overall equation

$6\ CO_2 + 6\ H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6\ O_2$

Limiting factors

The rate of photosynthesis is limited by light intensity, carbon dioxide concentration and temperature.